

so that the shift-register holds an undistorted signal, and finally estimating the new clock period, and so on.

For input signals with noise, this system will be much more sensitive to noise than the delay-lock discriminator. In the delay-lock discriminator the delayed signal is taken noise-free, while in this period estimator both signals are noisy.

Another fact that makes this system more noise sensitive is that we work with derivative estimates.

But if the input signal has a zero-mean-value we may integrate it (obtaining also a periodic signal) to alleviate this latter problem.

The system was simulated using a PDP-11/10 computer with many different waveforms. With signals with only one local maximum and one local minimum per period, the locking range is $\frac{2}{3} < P < 2$ with $T_c = 1$.

V. CONCLUSIONS

A delay-lock period estimator was presented that was able to track periodic signals sampled at a variable rate. This system is based on a short time correlation of the first difference (derivative) of the input signal correlated with the difference between the input signal and the approximately one period delayed input signal. The estimated delay is corrected sample to sample with a simple algorithm.

Many simulations with different waveforms confirm the feasibility of the method.

The noise behavior of this system was not studied yet.

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Model for the Power Spectral Density of Quantization Noise

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Abstract—A model is given for the spectrum of the noise produced by passing a signal through a uniform quantizer. The model requires knowledge only of the amplitude distribution of the derivative of this signal. The model is compared with experimental results for a sinusoidal input signal and the sum of two sinusoids. Conditions are given under which the quantization noise spectrum is white.

I. INTRODUCTION

This correspondence deals with an analysis of the spectrum of the error introduced when passing a signal through a uniform quantizer. The topic is by no means a new one. One of the most fundamental contributions to it is the work by Bennett [1]. His basic result is that when the quantizer is sufficiently loaded, that is to say, when the signal attains an appreciable number of levels of the quantizer, then the errors can be regarded as noise with a white spectrum over the signal band. Since Bennett published his paper this result has been

used widely to model quantization errors both for analog and for digital signals [2], [3]. In general, the model is quite accurate. Recently, Stripad and Snyder [4], dealing with the case of stochastic input signals, gave necessary and sufficient conditions for the quantization errors to have a white spectrum. The conditions are stated in terms of the characteristic function of the input process.

For deterministic signals too, the white noise model is remarkably accurate provided the signal has sufficient variation. However, conditions as to the amount of variation that is necessary have never been stated hitherto. For example, Oppenheim and Schaffer [2, sec. 9.2] speak of "complicated signals," by which they mean that the signals are "sufficiently complex and the quantization steps sufficiently small so that the amplitude is likely to traverse many quantization steps in going from sample to sample." Everyone is aware of specific examples where the model breaks down (e.g., step functions [2], dc-input [5]), while also limit cycles in digital filters are caused by correlation in the quantization noise of a quantizer in a feedback loop [6]. In this correspondence we will derive a more refined model for the power spectrum of the quantization noise for a deterministic input signal. Based on this model it is possible to indicate under which conditions the noise spectrum will be white in the band of interest and, if these conditions are not satisfied, what kind of deviation can be expected. The model is derived for the case of analog signals. The discrete-time case is briefly discussed in Section IV. As examples, a sinusoidal input signal and the sum of two sinusoids will be discussed. This case is of practical importance since sinusoidal signals are frequently used for testing electronic circuits. In particular, the testing of A/D and D/A conversion systems can most conveniently be done by means of a sinusoidal signal because for these signals the most sophisticated measurement equipment is available. If the sinusoidal signal has sufficient variation then the quantization errors may be assumed to have a white spectrum. But in the analysis of oversampled systems and when determining the signal to quantization noise ratio as a function of the signal power, it is necessary to know for which amplitude levels and which frequencies this assumption is reasonable.

II. DERIVATION OF THE MODEL

The errors (quantization noise) caused by uniformly quantizing a signal can be considered as being generated by passing this signal through the nonlinear characteristic depicted in Fig. 1.

Throughout this correspondence it will be assumed that the amplitudes of all signals are normalized such that the quantization step size is equal to unity. The nonlinear function $q(x)$ is periodic with period 1 and has a Fourier series representation

$$q(x) = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{\sin 2\pi kx}{\pi k}. \quad (1)$$

Therefore, the quantization noise is given by

$$y(t) = q(x(t)) = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{\sin 2\pi kx(t)}{\pi k} \quad (2)$$

$$= \sum_{k=1}^{\infty} y_k(t). \quad (3)$$

This is a sum of phase-modulated signals $y_k(t)$, with phase

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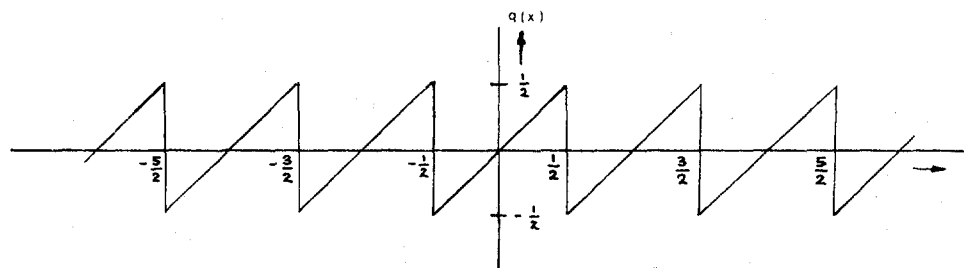


Fig. 1. Nonlinear characteristic that produces the quantization noise.

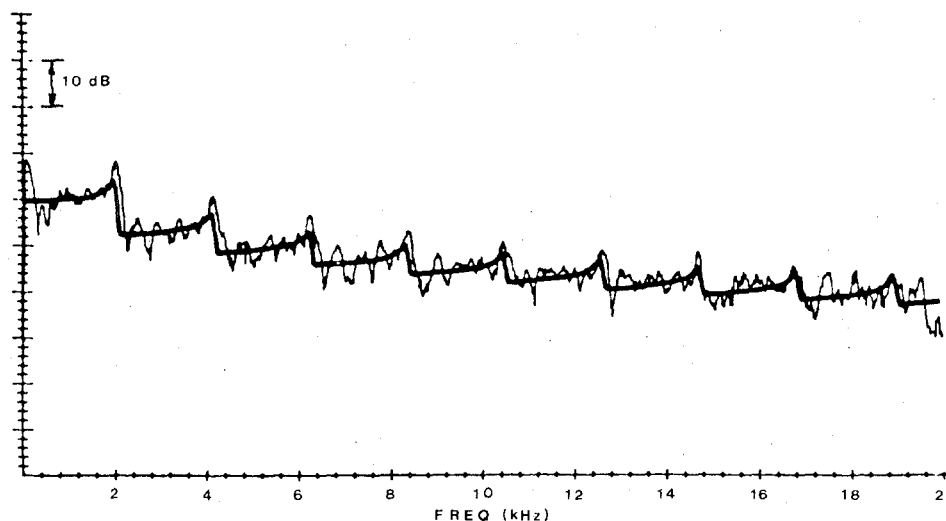


Fig. 2. Power spectral density of the quantization noise corresponding to a sinusoid of amplitude 15.5 and frequency 21.5 Hz, computed by the model of (5) and measured by a spectrum analyzer.

deviation $2\pi k$ and amplitude $1/\pi k$. Therefore, the spectrum of the quantization noise consists of the sum of spectra of phase-modulated signals. If $x(t)$ is a sinusoid, this spectrum can be determined analytically. The resulting form is rather complicated, however, and not very useful for describing the quantization noise spectrum [7]. If $x(t)$ is a more complicated signal, such an analytic expression cannot in general be found. We will, therefore, derive a model for the power spectrum of this noise that is somewhat simpler to deal with than the analytical formula. It should be noted that in the model it is not assumed that the signal or the noise are stochastic processes, but they may be deterministic signals.

To derive our model we use the following observations.

1) The power spectral density of a phase-modulated signal is proportional to the amplitude distribution of the derivative of the modulating signal [8], [9].

2) The spectral components of the various $y_k(t)$ add on a power basis. This latter assumption follows from the fact that the phases of the components at a certain frequency in $y_k(t)$ and $y_l(t)$ for $k \neq l$ do not show much correlation.

Denoting the amplitude distribution function (ADF) of the derivative of the input signal by $p_{\dot{x}}(a)$, the power spectral density of the k th component in (2) can be modeled according to [8] by

$$S_k(\omega) = \frac{1}{2\pi^2 k^3} p_{\dot{x}}(\omega/2\pi k). \quad (4)$$

Summing all components yields the power spectral density of the quantization noise as

$$S_y(\omega) = \frac{1}{2\pi^2} \sum_{k=1}^{\infty} p_{\dot{x}}(\omega/2\pi k)/k^3. \quad (5)$$

It will not always be easy to obtain expressions for the ADF $p_{\dot{x}}(a)$, but in some cases of interest this happens to be so.

III. QUANTIZATION NOISE FOR SINUSOIDAL SIGNALS

A particular example for which the ADF is easy to evaluate is the case of a single sinusoidal signal. The ADF of a sinusoid with amplitude A is given by

$$p(a; A) = \begin{cases} \frac{1}{A\pi} \cdot \frac{1}{\sqrt{1 - (a/A)^2}} & |a| < A \\ 0 & |a| > A. \end{cases} \quad (6)$$

Hence, if $x(t) = X \sin \omega_0 t$ we find $p_{\dot{x}}(a) = p(a; \omega_0 X)$. Therefore, the spectrum of the quantization noise of a sinusoidal signal is obtained by inserting (6) and $p_{\dot{x}}(a)$ into (5). The ADF in (6) has a pole at $a = A$. Therefore, the model of (5) predicts poles in the spectrum of the noise at frequencies $\omega = \omega_0 \cdot 2\pi kX$. In practice, due to the finite resolution of measurement equipment, the spectrum will be smoothed so that only finite contributions are found.

An example is given in Fig. 2, where the quantization noise spectrum is depicted for a sinusoid with amplitude $X = 15.5$ (relative to the quantization step size) and frequency $\omega_0/2\pi = 21.5$ Hz. The figure gives both the spectrum measured by means of a spectrum analyzer, using a filter with a bandwidth of 100 Hz, and the spectrum as predicted by the model ac-

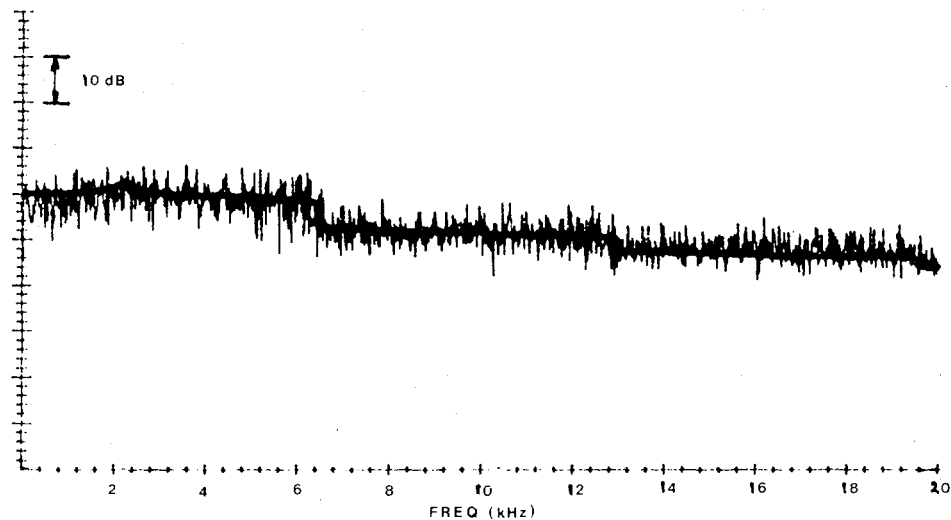


Fig. 3. Power spectral density of the quantization noise produced by a sum of two sinusoids with amplitudes 16 and 15 and frequencies 21.5 Hz and 45.9 Hz, respectively.

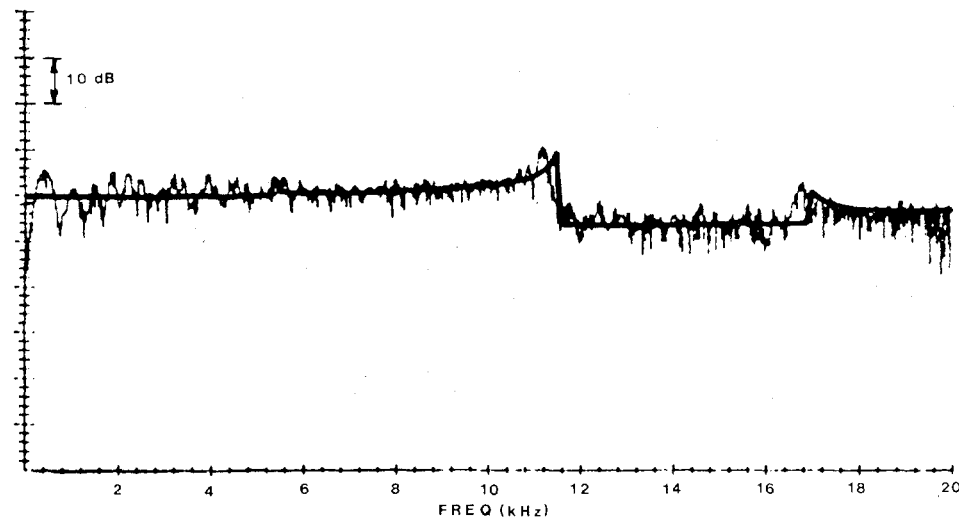


Fig. 4. Power spectral density of the quantization noise of a digital sinusoid of amplitude 94 and frequency 21.5 Hz, with a sampling rate of 40 kHz.

cording to (5), but taking into account the 100 Hz bandwidth of the filter by integrating $S_y(\omega)$ over a bandwidth of 100 Hz. It can be concluded from Fig. 2 that the model is in very good agreement with the measured spectrum even though the spectrum is by no means white.

It is clear that at low frequencies the spectrum is rather flat, but irregularities occur at the frequencies $\omega = \omega_0 \cdot 2\pi kX$.

Hence, to obtain a quantization noise spectrum that is flat in the band of interest we must restrict this band to

$$|\omega| \lesssim \pi X \omega_0. \quad (7)$$

This means that if our signal band is equal to 20 kHz, as is the case for audio signals, then the noise will be white within this band if the input signal obeys

$$X \cdot f_0 \gtrsim 6000 \text{ Hz}. \quad (8)$$

At an input frequency of 100 Hz this requires a minimum amplitude of 60 quantization steps.

A somewhat more complicated situation is that of a sum of two sinusoidal signals

$$x(t) = X_1 \sin \omega_1 t + X_2 \sin \omega_2 t. \quad (9)$$

Now use can be made of the fact that the ADF of a sum of

two sinusoidal signals is equal to the convolution of the ADF's of the individual components, provided that the frequencies are incommensurable. Hence, after differentiating (10) we find

$$p\dot{x}(a) = p(a; \omega_1 X_1) * p(a; \omega_2 X_2) \quad (10)$$

where $p(a; A)$ is given by (6).

Inserting (10) into (5) we find the corresponding power spectrum. An example is shown in Fig. 3, where $X_1 = 16$, $X_2 = 15$ (both normalized on the quantization step size), and $\omega_1/2\pi = 21.5$ Hz, $\omega_2/2\pi = 45.9$ Hz. Both a measured spectrum and a spectrum computed from the model are displayed in this figure. It is again seen that the model predicts the actual spectrum quite accurately. Due to the convolution operation the spectrum exhibits much less sharp peaks than in the previous case.

IV. DIGITAL SIGNALS

The issue of quantization most naturally arises when considering digital signals. Due to their representation by a finite number of bits, quantization is present in almost all digital signal processing systems and inherently occurs in the analog-to-digital conversion. Recalling that the sampling process is

associated with a folding of the corresponding spectrum around the sampling frequency [2], it is easy to extend the results of the previous section to the case of digital signals. Indeed, using the aliasing formula and assuming that the various components that fold back to the signal band add on a power basis, we get for the power spectrum of the digital quantization error

$$S_y^d(e^{j\omega t}) = \frac{1}{T^2} \sum_{m=-\infty}^{\infty} S_y\left(\omega + m \frac{2\pi}{T}\right) \quad (11)$$

where T is the sampling period.

As an example we may consider a digital sinusoid

$$x(n) = X \cos(n\omega_0 T). \quad (12)$$

Taking the sampling rate well below the first discontinuity in $S_y(\omega)$, i.e., in accordance with (7), the spectrum of the quantization noise will be white. If, however, the sampling rate is taken appreciably larger than this value, a nonwhite spectrum results, as can be seen from Fig. 4 for the case of a sinusoid with amplitude $X = 94$ and frequency $\omega_0/2\pi = 21.5$ Hz, sampled at a rate of 40 kHz. It can be seen from this figure that the model accurately predicts the quantization noise spectrum for this digital signal.

V. CONCLUSIONS

A model is introduced for the power spectral density of the quantization noise of a uniformly quantized input signal. In general, when the amplitude distribution of the derivative of the input signal is easy to obtain, then the model of the power spectral density of the quantization noise has a rather simple expression. We have shown that this model is quite accurate in the case of a single quantized sinusoid and for a quantized sum of two sinusoids. Simple expressions are derived from the model which give the conditions under which the quantization noise spectrum may be considered white in the band of interest. The model confirms the well-known fact that the quantization noise spectrum will be white in case the input signal to the quantizer has sufficient variation. Moreover, simple expressions are derived from the model which quantify this hitherto vague notion, in that they provide conditions for the quantization noise spectrum to be white in the band of interest.

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Some Notes on Closed Phase Glottal Inverse Filtering

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Abstract—Berouti *et al.* [2]–[4] and Wong *et al.* [5] have reported procedures for the "closed phase" identification of the all-pole model of the vocal tract. These procedures are shown to be equivalent in a certain case. We argue for the superior performance of Wong's method when modeling errors exist. Finally, a modification to Berouti's method is suggested.

INTRODUCTION

In the classic all-pole model of speech production [1] the vocal tract is represented by a difference equation of the form

$$x(n) = a_1 x(n-1) + a_2 x(n-2) + \cdots + a_M x(n-M) + b_0 v(n) \quad (1)$$

where $v(n)$ is the glottal volume velocity and $x(n)$ the volume velocity just prior to the lips.¹ For economy of space we will often set $M = 2$ in this correspondence, although typically M will equal 6–10 in speech processing.

Berouti *et al.* [2]–[4], and more recently, Wong *et al.* [5] have reported procedures for identification of the coefficient set $\{a_i\}$ based on the assumption of a "closed phase" (CP) in the glottal cycle. During the CP, the sequence $v(n)$ takes an unknown² constant value c .

In Berouti's method $\{a_i\}$ is identified by solving the simultaneous equations

$$\begin{aligned} x(n_0) - \hat{a}_1 x(n_0 - 1) - \hat{a}_2 x(n_0 - 2) &= \hat{b}_0 \hat{c} \\ x(n_0 + 1) - \hat{a}_1 x(n_0) - \hat{a}_2 x(n_0 - 1) &= \hat{b}_0 \hat{c} \\ x(n_0 + 2) - \hat{a}_1 x(n_0 + 1) - \hat{a}_2 x(n_0) &= \hat{b}_0 \hat{c} \end{aligned} \quad (2)$$

during the CP for $\{\hat{a}_i\}$, where $\hat{b}_0$ is treated as an unknown and $\hat{c}$ is chosen arbitrarily. For reasons discussed below, we suggest modifying the procedure so that the preemphasized speech, say $y(n)$, rather than the deemphasized $x(n)$ is used in the solution

$$\begin{aligned} y(n_0) - \hat{a}_1 y(n_0 - 1) - \hat{a}_2 y(n_0 - 2) &= 0 \\ y(n_0 + 1) - \hat{a}_1 y(n_0) - \hat{a}_2 y(n_0 - 1) &= 0. \end{aligned} \quad (3)$$

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¹It is assumed here that the speech signal, say $s(n)$, has been preprocessed to invert the radiation effect of the lips; see, e.g., [1], i.e.,

$$X(z) = L^{-1}(z) S(z)$$

where

$$L^{-1}(z) = \frac{1}{1 - \alpha z^{-1}}$$

$$\alpha \approx 1$$

$L(z)$ is the impedance between the volume velocity at the lips $x(n)$ and the speech pressure waveform $s(n)$.

²The equipment used to record the speech will generally remove its average value so that the recorded waveform will be shifted "downward," with respect to the unrecorded, by the dc value of the unrecorded signal. If and only if there is zero flow during the CP at the glottal level can c be interpreted as the negative of the dc value of the unrecorded voice signal, and b_0 the dc gain constant of the vocal tract.